

GIS & Visualization Layer (Layer 5)

Technical Architecture & Spatial Rendering Specification

DOCUMENT ID SUBSENSE-TDD-GIS-005	SYSTEM LAYER Layer 5: Spatial & GIS Rendering Stack	TARGET RUNTIME PostGIS / MapLibre GL / CesiumJS	COMPLIANCE STANDARD DGMS Geotechnical GIS Guidelines
RELEASE VERSION v2.1 (Production Architecture)	LEAD AUTHOR SubSense Spatial Computing Team	CLASSIFICATION Technical Architecture Specification	REVIEW STATUS Approved for Implementation

1. PURPOSE & SCOPE

The **GIS & Visualization Layer** is the spatial rendering and situational-awareness component of **SubSense**. Positioned at *Software Stack Layer 5*, it takes the risk scores, deformation progression forecasts, Kriging surfaces, and graph correlation outputs produced by the upstream *AI/ML Intelligence Layer (Layer 4)* and converts them into map-based, spatially-grounded visual products. These visual outputs allow mine operators, safety inspectors, geotechnical planners, and regulatory bodies (such as DGMS) to take direct preventative action without having to interpret raw time-series or multi-dimensional tensor arrays.

This document specifies the technical design, architectural pipelines, data schemas, coordinate systems, and serving infrastructure across the layer's six core sub-components:

- **Live GIS Deformation Heatmap (5.1):** Real-time raster generation with variance overlay.
 - **Risk-Zone Boundary Engine (5.2):** Contour extraction and stable polygon identifier tracking.
 - **3D Subsurface Digital Twin (5.3):** Integrated mine geometry + draped terrain deformation.
- **Sentinel-1 InSAR Fusion Overlay (5.4):** Macro-scale satellite cross-check visualization.
 - **Map Tiling & Delivery Architecture (5.5):** Multi-tenant 2D tile pyramids & 3D asset caching.
 - **Historical Replay & What-If Simulation (5.6):** Temporal scrubbing and hypothetical panel analysis.

2. CORE ARCHITECTURAL DESIGN PRINCIPLES

1. Model Output In, Map Layer Out

Layer 5 does not compute geomechanical risk. It consumes the continuous risk surfaces, zone boundaries, and explainability payloads generated by Layer 4, maintaining strict separation of concerns focused entirely on spatial rendering, layering, and performant interaction.

2. Multi-Resolution by Design

Ground-sensor data is dense, highly accurate, and streams in near-real-time; satellite InSAR data is coarse (20m pixels) and infrequent (6–12 days). The layer renders both at their native spatio-temporal fidelity without forcing artificial synchronization.

3. Progressive Disclosure UI/UX

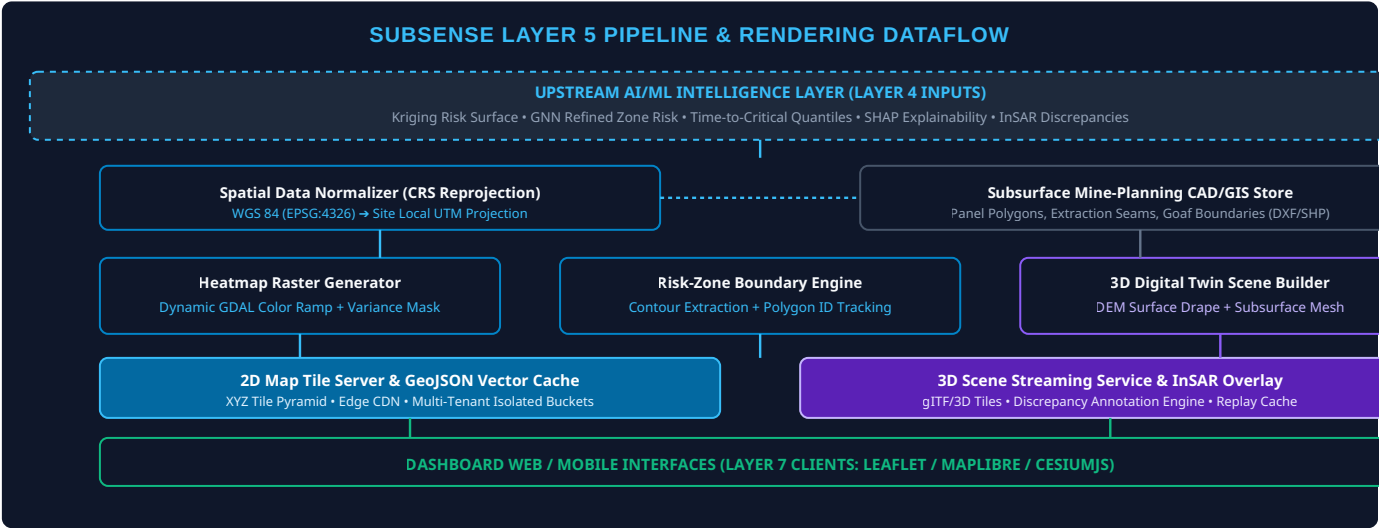
The primary operational screen remains uncluttered during an emergency (displaying only the live heatmap and active risk zones). Deeper diagnostic layers (3D digital twin, InSAR discrepancies, SHAP feature attributions) are opt-in overlays to avoid visual cognitive overload.

4. Unified Spatial Reference & Degraded Operation

All GPS node markers, CAD mine-workings, and satellite interferograms resolve through a shared Spatial Data Normalizer to prevent coordinate drift. Under field backhaul failure, cached map layers render gracefully with unambiguous staleness badges.

3. HIGH-LEVEL GIS SYSTEM ARCHITECTURE

The GIS & Visualization pipeline ingests upstream raster/vector packages, transforms them through a centralized Coordinate Reference System normalizer, and distributes lightweight slippy-map tiles and 3D terrain assets to multi-tenant client interfaces.



4. COORDINATE REFERENCE SYSTEM & SPATIAL DATA MODEL

Mining operations rely on legacy underground survey grids, CAD mine-workings, and global GPS networks. SubSense guarantees sub-meter geometric alignment via a standardized spatial pipeline:

- **Ingestion & GPS Base:** All wireless mesh nodes transmit coordinates in standard **WGS 84 (EPSG:4326)**.
- **Planar Spatial Computations:** For accurate metric distance, Kriging interpolation, and polygon buffering, coordinates are reprojected by the *Spatial Data Normalizer* to site-specific **Universal Transverse Mercator (UTM)** zones (e.g., WGS 84 / UTM Zone 45N, EPSG: 32645 for Jharkhand/Raniganj coalfields).
- **Subsurface CAD/Mine Georeferencing:** Underground mine layout drawings (supplied in arbitrary local mine coordinates or DXF grids) undergo an affine georeferencing transformation ($X_{UTM} = A \cdot x_{local} + B \cdot y_{local} + C$) calibrated using permanent surface shaft tie-points. Calibration matrices are stored per mine site in PostGIS.

5. COMPONENT TECHNICAL SPECIFICATIONS

5.1 Live GIS Deformation Heatmap Rendering

CORE 2D RASTER

Goal: Render an uninterrupted, continuously updating deformation heatmap across the mine concession for immediate hazard assessment.

Approach: Ingests the continuous 2D Kriging risk raster from Layer 4. A GDAL-based pipeline applies a calibrated, colorblind-safe color ramp (Low: #22c55e green, Advisory: #eab308 yellow, Warning: #f97316 orange, Critical: #ef4444 red). Alongside deformation magnitude, the Kriging estimation variance (σ_K^2) is processed into a stippled or opacity-modulated confidence mask, highlighting regions where sparse node coverage requires cautious interpretation. Outdated gateway zones display diagonal watermark striping indicating data staleness.

Inputs:	Kriging risk surface GeoTIFF, Kriging estimation variance grid, node health metadata.
Outputs:	Web map raster tile pyramid (XYZ scheme), tooltip hover payload bound to sensor positions.
Refresh Cadence:	Near-real-time ($\leq 30s$), tied directly to Layer 4 gateway sync cycles.

5.2 Risk-Zone Boundary Generation & Overlay

VECTOR GEOMETRY

Goal: Convert raw continuous heatmaps into discrete, legally accountable, named risk-zone polygons that can be referenced in statutory evacuation orders and safety circulars.

Approach: Extracts contour isolines from the risk raster matching Fusion Engine alert thresholds (Advisory ≥ 0.65 , Warning ≥ 0.75 , Critical ≥ 0.85). Polygons are smoothed using Visvalingam-Whyatt simplification, filtered with a minimum bounding area threshold ($\geq 250\text{ m}^2$) to eliminate sensor noise micro-islands, and validated for strict polygon topology (no self-intersections). Polygons retain stable naming identifiers (e.g., "Zone-Panel7-B") across refresh cycles by calculating maximum Jaccard spatial intersection with prior-cycle geometry.

Inputs:	Continuous risk raster, alert classification thresholds, historical zone cache.
Outputs:	GeoJSON FeatureCollection with attached properties (severity, TTC window, contributing nodes, SHAP narrative).
Refresh Cadence:	Synchronized with the heatmap generation cycle.

5.3 3D Subsurface Digital Twin

WEBGL / CESIUMJS

Goal: Provide an integrated 3D spatial twin linking surface subsidence flexure directly with underground bord-and-pillar or longwall goaf extraction geometry.

Approach: The subsurface model imports 3D underground extraction workings (galleries, coal pillars, goaf voids, and overburden lithology) rendered as translucent volumetric meshes. The surface terrain Digital Elevation Model (DEM) is dynamically draped with the live deformation heatmap. Real-world subsidence (millimeter-to-centimeter scale) is imperceptible over square-kilometer concessions; the twin incorporates an adjustable vertical displacement exaggeration scale (e.g., 10× to 50×) with an unambiguous on-screen watermark to ensure realistic comprehension.

Inputs:	Subsurface CAD extraction meshes, surface elevation DEM (1m–5m resolution), live risk texture raster, 3D node coordinates.
Outputs:	Interactive 3D Tiles / glTF scene streamed to WebGL clients, high-resolution static rendering export for DGMS reports.
Refresh Cadence:	Surface texture updates every ingestion cycle; subsurface mesh updates upon formal mine-plan survey revision.

5.4 Sentinel-1 InSAR Macro-Fusion Visualization

GEODETIC CROSS-CHECK

Goal: Visually correlate regional satellite interferometric ground-displacement basins with local high-frequency ground mesh sensor readings.

Approach: InSAR Line-of-Sight (LOS) velocity displacement maps (derived from Copernicus Sentinel-1 SLC interferograms) are displayed using a distinct hatched vector overlay to prevent confusion with real-time sensor heatmaps. Discrepancy flags emitted by Layer 4 (Section 5.10) render as visual alert beacons, indicating unmonitored subsidence bowls outside the active mesh or uncorroborated local sensor anomalies requiring on-site field verification.

Inputs:	Sentinel-1 InSAR LOS displacement raster, geocoded discrepancy feature list from Layer 4.
Outputs:	Toggable 2D/3D InSAR layer, geolocated discrepancy callout cards with acquisition date badges.
Refresh Cadence:	Asynchronous, matching satellite pass revisit frequency (typically 6 to 12 days).

5.5 Map Tiling, Caching & Multi-Tenant Delivery Architecture

SERVING INFRASTRUCTURE

Goal: Deliver sub-second map pan/zoom responsiveness across desktop control rooms and field tablets operating over constrained rural 4G or satellite backhauls.

Approach: Generated heatmaps are sliced into standard XYZ tile pyramids (zoom levels 12 to 18) and cached within a distributed edge CDN. Vector polygons and node states stream via lightweight GeoJSON over HTTP/2. The multi-tenant architecture strictly isolates tenant data at the PostgreSQL row level and S3 tile bucket path (/tiles/{tenant_id}/{site_id}/{z}/{x}/{y}.png). Under network dropouts, progressive local caching displays the last valid state stamped with an prominent amber staleness countdown.

Inputs:	Compiled raster grids, vector zone layers, client spatial viewport requests.
Outputs:	Standard XYZ slippy-map raster tiles, cached vector GeoJSON endpoints, 3D tile binary streams.
Performance:	≤ 250 ms tile retrieval latency at 95th percentile under concurrent multi-user load.

5.6 Historical Replay & What-If Scenario Rendering

SIMULATION & AUDIT

Goal: Provide chronological post-incident audit replay and predictive what-if extraction planning visualizations for mine engineers.

Approach: Historical raster and vector states are snapshotted and indexed by timestamp in PostGIS. The dashboard client features a VCR-style timeline scrubber enabling arbitrary playback of past strata movements. In "What-If" planning mode, engineers load proposed longwall panel extraction boundaries; the system renders simulated surface deformation profiles calculated from empirical subsidence models (e.g., Peck or empirical NCB profiles), enclosed within a high-contrast striped border confirming non-live simulation status.

Inputs:	Time-indexed historical state archive; CAD proposed extraction vectors (for simulation).
Outputs:	Time-scrubbable map sequence, watermarked simulation scene with side-by-side risk deltas.

6. FORMAL DATA CONTRACTS & API SCHEMAS

Standardized JSON contracts define communications between Layer 4 analytics, the spatial store, and client viewports:

6.1 HEATMAP TILE REQUEST & METADATA RESPONSE HEADER

```
{
  "site_id": "PANEL7-JHARIA",
  "layer": "risk_deformation_heatmap",
  "tile_coordinates": { "z": 16, "x": 46962, "y": 30468 },
  "as_of": "2026-09-09T04:15:00.000Z",
  "data_currency_seconds": 18,
  "is_stale": false,
  "kriging_variance_included": true
}
```

6.2 RISK-ZONE POLYGON FEATURE SCHEMA (GEOJSON OUTPUT)

```
{
  "type": "Feature",
  "id": "ZONE-PANEL7-C",
  "properties": {
    "site_id": "PANEL7-JHARIA",
    "zone_name": "Goaf Abutment Zone C",
    "severity_tier": "warning",
    "time_to_critical_hours": [6.0, 14.0],
    "model_confidence": 0.79,
    "affected_node_ids": ["SS-PANEL7-N042", "SS-PANEL7-N043", "SS-PANEL7-N051"],
    "primary_contributing_sensors": ["tilt_deg", "displacement_mm"],
    "explanation_summary": "Sustained tilt increase at N042, corroborated by 3 neighboring nodes over 40 min.",
    "area_sq_meters": 18450.2,
    "centroid_gps": { "lat": 23.7915, "lon": 86.4335 },
    "last_updated": "2026-09-09T04:15:00.000Z"
  },
  "geometry": {
    "type": "Polygon",
    "coordinates": [[[86.4328, 23.7910], [86.4342, 23.7911], [86.4340, 23.7922], [86.4325, 23.7920], [86.4328, 23.7910]]]
  }
}
```

6.3 INSAR DISCREPANCY ANNOTATION MARKER SCHEMA

```
{
  "annotation_id": "INSAR-DISC-2026-088",
  "site_id": "PANEL7-JHARIA",
  "location": { "lat": 23.792015, "lon": 86.434020 },
  "insar_acquisition_date": "2026-08-28",
  "discrepancy_type": "satellite_motion_unmonitored_by_ground_mesh",
  "los_velocity_mm_year": -34.5,
  "description": "InSAR indicates 14mm subsidence basin outside active sensor array. Geotechnical review required.",
  "recommended_action": "Relocate wireless sensor nodes N088 and N089 120m northeast."
}
```

7. TECHNOLOGY STACK SUMMARY

COMPONENT	SELECTED PRODUCTION TECHNOLOGY	ARCHITECTURAL RATIONALE
2D Web Map Client	MapLibre GL JS / Leaflet	Hardware-accelerated vector tile rendering with offline raster fallbacks for mobile field tablets.
Raster Tile Generation	GDAL / Rasterio + Custom Python XYZ Worker	High-throughput generation of dynamic, color-mapped Web Mercator tiles directly from Kriging matrices.
Spatial Database & Store	PostGIS (PostgreSQL 16)	Standard-compliant spatial indexing (R-Tree / GiST), temporal partitioning, and polygon intersection logic.
3D Digital Twin Engine	CesiumJS (WebGL / WebGPU)	Accurate global ellipsoid projection, native 3D terrain draping, and glTF underground mesh streaming.
CAD/GIS Ingestion	Fiona, Shapely, ezdxf	Parses statutory mine extraction maps from AutoCAD DXF, ESRI Shapefile, and LandXML formats.
Satellite InSAR Ingestion	ESA SNAP Engine, Sentinel-Hub API	Automated ingestion of interferometric deformation rasters and spatial alignment with surface grids.
Tile Caching & CDN	Redis + Cloudflare / MinIO Object Store	In-memory tile caching for active zones, partitioned by tenant ID to guarantee zero data leakage.

8. PERFORMANCE & RELIABILITY TARGETS

Performance Indicator	Target Threshold	Operational Context & Verification Protocol
End-to-End Tile Rendering SLA	≤ 30 Seconds Post-Ingestion	Total duration from gateway telemetry arrival to newly published heatmap tiles visible on dashboard.
2D Tile Fetch Latency	≤ 250 ms (95th Percentile)	Tile server response time under 100 concurrent operator sessions across control rooms.
Zone ID Persistence Stability	≥ 98.5% Overlap Consistency	Eliminates confusing polygon renaming across consecutive cycles to maintain communication continuity.
3D Scene Framerate	≥ 45 FPS (Field Hardware)	Smooth interaction verified on standard Core i5 / 8GB RAM field laptops without dedicated GPU acceleration.
Multi-Tenant Data Isolation	100% Partition Verification	Cryptographic validation that no tenant can query or fetch tile assets belonging to another mine operator.

9. OPEN ENGINEERING CONSIDERATIONS & RISK MITIGATION

Implementation Considerations & Geotechnical Mitigations

- Vertical Exaggeration Calibration:** At true scale, 15mm of subsidence across a 500m panel is invisible on a 3D monitor. *Mitigation:* Apply an adaptive 25× vertical displacement exaggeration with a permanent on-screen scale ruler to prevent operators misreading ground distortion.
- Graceful Degradation for Missing CAD Workings:** Older mining concessions often lack digitized 3D CAD panel drawings. *Mitigation:* The digital twin engine features an automatic fallback to a surface-only 2.5D terrain drape when underground geometry is unavailable.
- Contour Noise Suppression:** Sparse sensor setups can cause Kriging contours to break into erratic, small polygons. *Mitigation:* Implement a morphological closing filter and strict 250 m² area cutoff before assigning formal zone identifiers.
- InSAR vs. Ground Sensor Misinterpretation:** InSAR represents 12-day historical movement, whereas ground sensors represent real-time kinematic activity. *Mitigation:* Separate the layers with distinctly styled hatch patterns and unambiguous temporal warning tags.